Nathan Walker, BS Electrical Engineering

Cellphone: 385-775-6664 **nate.walker.timp@gmail.com**

Executive Summary

Electrical engineer with experience designing, integrating, and testing embedded systems for sensing and aerospace applications. Developed real-time signal processing algorithms, contributed to a rocket attitude control system through sensor integration and control validation, and built FPGA-based hardware/software solutions for custom processing platforms. Skilled in system integration, debugging, and iterative hardware testing, with a passion for developing reliable avionics and embedded electronics for aerospace applications.

Work Experience

Graduate Researcher May 2024 – Present

Thesis: *Design and Implementation of Adaptive Active Snore Reduction Using Near-Field Array Processing*

- Developed real-time embedded signal processing algorithms on Linux, implementing adaptive filtering and array processing for noise cancellation.
- Repaired and debugged custom electronics designed by others with no circuit documentation, identifying faults and restoring full functionality.
- Used MATLAB and C to analyze data, supporting debugging and performance optimization.
- Created custom PCB enclosures in Solid Edge to support mechanical stability and system function
- Effectively communicated complex technical problems and solutions to advising professors.
- Ensured data integrity during collection and analysis of automated experimental measurements.

ASPIRE Research Assistant May 2023 – April 2024

- Selected for the ASPIRE research program based on academic performance and faculty recommendation.
- Designed and executed PCB and component testing protocols, including soldering, signal verification, and measurement with lab instruments, ensuring accurate experimental results.
- Developed hardware solutions for research projects under size, cost, and performance constraints.
- Assembled complex PCBs and coordinated multi-component research projects from design to execution, demonstrating hands-on troubleshooting and project management.

Engineering Projects

International Future Energy Challenge – Team Lead July 2023 – July 2024

- Led a team of 3 undergraduates, taking primary responsibility for hardware design and component selection while delegating tasks to deliver a high-efficiency Class D amplifier.
- Designed and implemented custom magnetic components using Ansys simulations to meet performance goals.
- Created and assembled multiple PCBs using Altium, improving product quality and testability.
- Troubleshot and debugged custom PCBs under tight design constraints, applying circuit analysis, measurement techniques, and hands-on problem-solving.

Skills

Altium/KiCad	C/C++ & Python	MATLAB/Data Analysis
Linux/GitHub/Scripting	Baking	Signal Processing
Communication & Teamwork	Hardware Troubleshooting	VHDL/Verilog

Education

Utah State University – Logan Utah

- M.S. Electrical Engineering – 3.83 GPA May 2026
- B.S. Electrical Engineering – 3.97 GPA May 2025